

WEAVE: Breaking the Identity Shortcut in Lightweight Style Transfer with Wavelets

Anonymous ICME submission

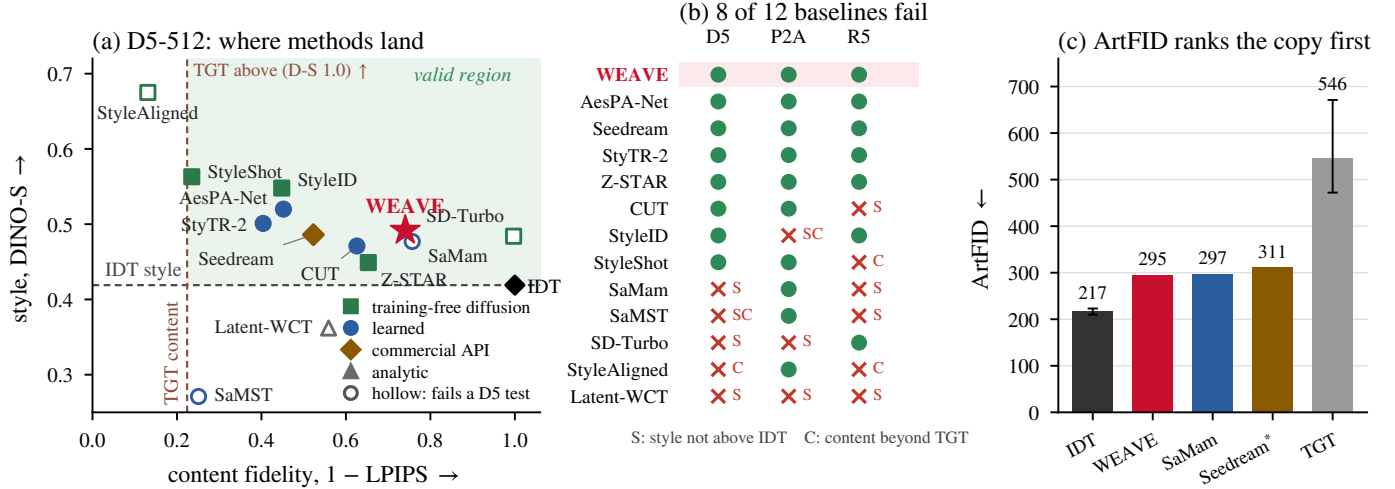


Fig. 1. The identity shortcut, exposed. IDT returns the source; TGT returns a reference image of the target style. (a) On D5-512, valid transfer lies above IDT in style and right of TGT in content; hollow markers fail at least one test, e.g., SD-Turbo and SaMam score no higher than a copy in CLIP-S. (b) Eight of twelve baselines fail on at least one benchmark; WEAVE passes all three. (c) ArtFID ranks the unchanged input first (*Seedream: 720 of 750 requests; intervals: bootstrap for IDT, 30 reference draws for TGT).

Abstract—Lightweight style transfer is built for consumer devices, yet compact models routinely return images no more stylized than an unchanged copy of the input, and standard content metrics reward this identity shortcut. Two zero-cost references, the source and a reference image of the target style, bracket valid transfer and expose it: eight of twelve baselines leave the bracket on at least one of three benchmarks. The cause is spectral. In latent velocity matching, coarse structure carries most of the gradient energy yet is the least style-discriminative band. WEAVE removes this imbalance in the Haar subbands of a frozen autoencoder latent: a source-anchored target provably suppresses structural motion, a compact network learns band-weighted texture transport, and stepwise statistical alignment injects high-frequency style. Trained in 1.4 minutes on one consumer GPU with 1.04M parameters, WEAVE is valid on every benchmark and Pareto-optimal against all baselines at 512 pixels.

Index Terms—style transfer, flow matching, wavelet decomposition, evaluation protocol, efficient generation

I. INTRODUCTION

Style transfer now runs in camera apps, creative tools, and on-device media pipelines, where memory and latency are tight. The strongest results come from diffusion editors and image foundation models that run a billion-parameter denoiser per image, some needing more than 60 GB of GPU memory [1], [2]. Compact models fit the budget [3]–[5], but do they actually change the style? We show that they often do not, and we build one that does.

The identity shortcut. Simply copying the input scores a CLIP-S of 0.693 on our main benchmark, beating two recent compact models (SaMam 0.582, SaMST 0.618). ArtFID [6] ranks the copy first (Fig. 1c), and LPIPS [7] declares it perfect. We therefore report two zero-cost outputs with every metric: IDT returns the source, and TGT returns a reference image of the target style. A valid transfer must beat IDT on style and stay closer to the source than TGT on content. Under this IDT–TGT sandwich, eight of twelve baselines fail on at least one benchmark, and every compact baseline (under 10M trainable parameters) is, somewhere, no more stylized than a copy (Fig. 1b).

Why compact models take the shortcut. The cause is the training signal, not capacity. Style transfer must keep structure while changing texture, yet a velocity field regressed on the whole latent scores both with one loss. In Haar coordinates, the coarse LL band carries 69.5% of the gradient energy across 5,000 cross-style pairs while being the least style-discriminative band: its between-/within-style variance ratio is 0.12, against 0.56 for the finest band HH (Fig. 2). This imbalance is a property of the regression target, so widening the network leaves it intact. Networks that fit low frequencies first [8], [9] then settle near the source, while large editors escape only by overwriting the image.

WEAVE resolves the conflict by giving structure and style their own subbands of a frozen SD1.5 VAE latent [10]. A source-anchored endpoint restricts the requested LL motion to

a channel-wise affine change of the source, which provably removes at least 91% of its energy and cuts LL’s share from 69.5% to at most 17%. A 1.04M-parameter velocity network learns band-weighted transport, steered by pooled, coordinate-free codes of the reference, and an AdaIN update aligns high-frequency statistics after every Euler step. Training takes 82.8 s on one RTX 3060 and stops itself at an internal gradient transition. WEAVE is the only compact model valid on all three benchmarks, is Pareto-optimal against all twelve baselines at 512 px, and is $11\times$ smaller and at least $11\times$ faster than every other valid learned model.

Contributions. (1) *Evaluation*: the IDT-TGT sandwich, a zero-cost, metric-agnostic validity test that exposes the identity shortcut and content collapse, and an audit of twelve methods on three benchmarks. (2) *Analysis*: a subband measurement and a bound that tie the shortcut to low-frequency gradient dominance and dictate the transport target. (3) *Method*: wavelet-decoupled velocity matching with orientation-specific texture conditioning, stepwise high-frequency alignment, and metric-free stopping. (4) *Results*: the only compact model valid everywhere, Pareto-optimal at 512 px, with 1.04M parameters and 1.4 training minutes, plus a training-free plug-in that improves a frozen SD1.5 editor.

II. RELATED WORK

Arbitrary and lightweight style transfer. Gatys et al. optimized Gram statistics [11], [12]; AdaIN and WCT made transfer feed-forward [13], [14]. SANet, StyTr², and AesPA-Net add attention [15]–[17], ArtFlow avoids content leak with reversible flows [18], and CUT learns unpaired translation by patch contrast [19]. MicroAST uses micro encoders and a micro decoder [3], and Mamba-based SaMST and SaMam reduce sequence cost [4], [5], [20]. Frequency cues appear in FDD, which mixes content and style spectra [21], in AesFA, which decomposes features by frequency [22], and in the wavelet-corrected WCT² [23]. These methods separate frequencies inside a feed-forward decoder; WEAVE makes Haar subbands the coordinates of the transport objective itself, with per-band targets, weights, and updates dictated by a measured imbalance.

Diffusion editors and fast translation. Latent diffusion [10] enables SDEdit, Prompt-to-Prompt, StyleID, StyleAligned, InST, Z-STAR, IP-Adapter, and StyleShot [24]–[31]. SD-Turbo distills sampling to one step [32], and LBM performs one-step latent bridge matching [33]. All of them run a large backbone for every image; WEAVE keeps only the frozen VAE, and its Haar operator even improves such an editor without retraining.

Flow matching and frequency-aware generation. Rectified flow and flow matching [34], [35] regress velocities along endpoint interpolations, and Schrödinger bridges [36]–[39] handle unpaired domains. FDIT constrains frequency bands separately for translation [40], FreqFlow reweights frequencies for class-conditional generation [41], and wavelet diffusion accelerates sampling [42]. WEAVE changes the endpoint, basis, and band weights of the regression itself.

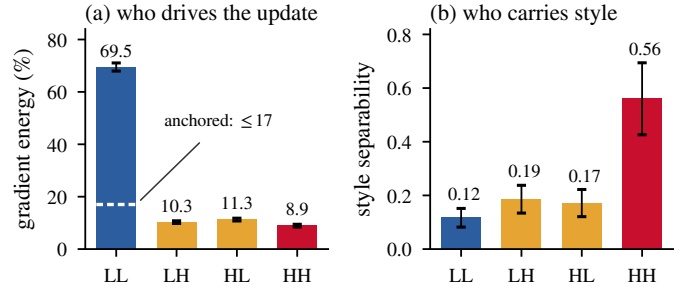


Fig. 2. Diagnosis on 5,000 cross-style D5 latent pairs. (a) Share of gradient energy per Haar band under the direct target; the dashed line marks the bound for the source-anchored target (Proposition 1). (b) Between-/within-style variance ratio of per-image band statistics. Error bars: standard error over the 20 style routes (a) and over styles (b).

Evaluation. LPIPS [7] and ArtFID [6] are standard, yet neither penalizes a copy, and a recent review finds no consensus on style-transfer evaluation [43]. The sandwich gives every metric a direction at zero cost.

III. DIAGNOSING THE IDENTITY SHORTCUT

A. The IDT-TGT Sandwich

A request consists of a source image x and a target style s , represented by a reference image r_s . We set $y_{\text{IDT}} = x$ and $y_{\text{TGT}} = r_s$ and call a method *valid* on a benchmark if its benchmark means satisfy

$$S(y) > S(y_{\text{IDT}}) \quad \forall S \in \mathcal{S}, \quad C(y) \succ C(y_{\text{TGT}}) \quad \forall C \in \mathcal{C}, \quad (1)$$

where $\mathcal{S} = \{\text{DINO-S}, \text{CLIP-S}\}$, $\mathcal{C} = \{\text{LPIPS}, \text{DINO-C}\}$, and $\succ$ means closer to the source. The first condition rules out the identity shortcut, the second rules out content collapse. The test costs nothing, works in raw metric units, needs no weights or thresholds, and applies to any style and content metric. Copying is a deceptively strong baseline because content images already share features with any art collection: on D5, IDT scores DINO-S 0.419 and CLIP-S 0.693.

B. Spectral Dominance in Latent Velocity Matching

Let $\mathcal{W}(z) = (\ell, h_1, h_2, h_3)$ be the one-level orthonormal Haar transform of a latent $z \in \mathbb{R}^{4 \times H \times W}$, with LL band ℓ and the oriented bands LH, HL, and HH. Flow matching [34], [35] regresses a velocity v onto the displacement $u = z_1 - z_0$ between endpoints. By Parseval, $\|v - u\|^2 = \sum_b \|v_b - u_b\|^2$: the loss splits into independent band terms, and at initialization the share of band b in the gradient energy equals its share of the displacement energy, $\|u_b\|^2 / \|u\|^2$. Fig. 2 measures both on 5,000 cross-style D5 pairs with the direct target $z_1 = z_s$. LL holds 69.5% of the gradient energy, yet its style separability, the between-/within-style variance ratio of per-image band statistics, is the lowest (0.12 versus 0.56 for HH). The objective is driven by the band that matters least for style, and moving ℓ_c toward ℓ_s asks the network to reproduce the unrelated layout of x_s , which it can neither predict nor should copy.

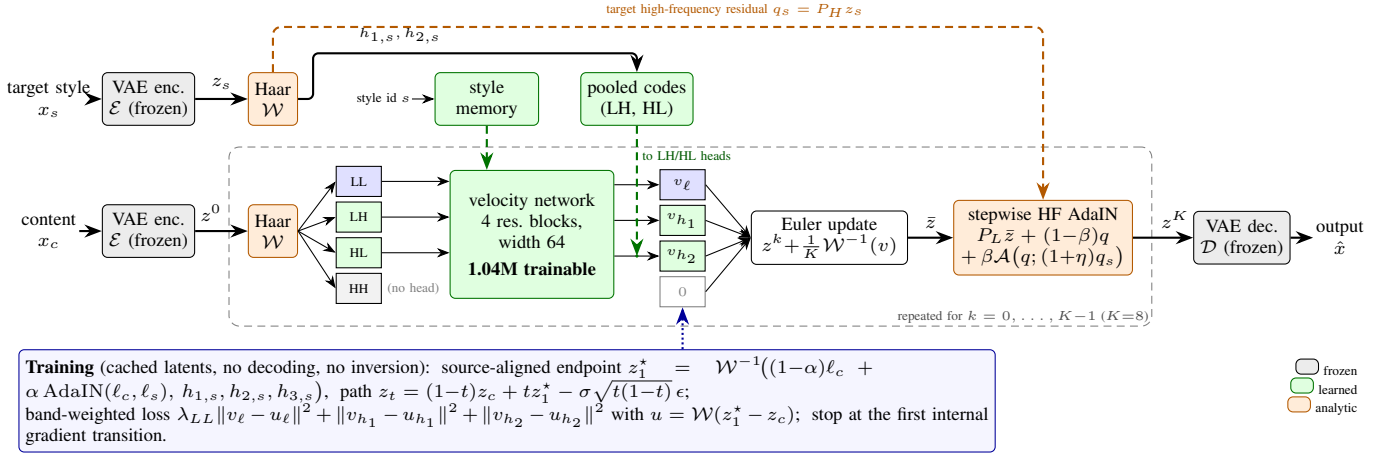


Fig. 3. WEAVE. A one-level Haar transform exposes LL and three high-frequency bands of the VAE latent. The compact network predicts LL/LH/HL velocities; pooled reference LH/HL codes modulate only the matching heads, and HH has no learned head. After every Euler update, AdaIN aligns the high-frequency residual with the reference statistics; LL is never altered by this step. Only the green blocks are trained.

IV. WEAVE

A. Setting

WEAVE is trained on unpaired image collections of K styles. A request names a target style by its index s and one reference image x_s ; at test time this is the first held-out image of the style, the same image used for TGT. We work on latents $z_c = \mathcal{E}(x_c)$ and $z_s = \mathcal{E}(x_s)$ of the frozen VAE, cached once before training, and write $P_L z = \mathcal{W}^{-1}(\ell, 0, 0, 0)$ and $P_H = I - P_L$ (Fig. 3).

B. Source-Anchored Endpoint

WEAVE anchors structure at the source and takes texture from the reference:

$$z_1^* = \mathcal{W}^{-1}((1-\alpha)\ell_c + \alpha \text{AdaIN}(\ell_c, \ell_s), h_{1,s}, h_{2,s}, h_{3,s}), \quad (2)$$

with $\alpha = 0.3$ and channel-wise spatial statistics in AdaIN [13].

Proposition 1: For one channel, let $\ell_c, \ell_s \in \mathbb{R}^N$ have means μ_c, μ_s , standard deviations σ_c, σ_s , and correlation ρ . The LL velocity requested by (2), $u_\ell = \alpha(\text{AdaIN}(\ell_c, \ell_s) - \ell_c)$, is an affine function of ℓ_c and

$$\|u_\ell\|^2 = \alpha^2 N[(\sigma_s - \sigma_c)^2 + (\mu_s - \mu_c)^2] \leq \alpha^2 \|\ell_s - \ell_c\|^2. \quad (3)$$

Proof: $\text{AdaIN}(\ell_c, \ell_s) - \ell_c = (\sigma_s/\sigma_c - 1)(\ell_c - \mu_c) + (\mu_s - \mu_c)$ gives the equality. Since $\|\ell_s - \ell_c\|^2 = N[\sigma_s^2 + \sigma_c^2 - 2\rho\sigma_s\sigma_c + (\mu_s - \mu_c)^2]$, the gap is $2N\sigma_s\sigma_c(1 - \rho) \geq 0$. ■

Two consequences follow. The requested structural motion only changes per-channel gain and offset, i.e., palette and contrast, and contains no layout of x_s . Because the high-frequency target is unchanged, averaging (3) over the pairs of Fig. 2 lowers LL's share of the displacement energy from 69.5% to at most 17.0%. Training samples $z_t = (1-t)z_c + tz_1^* - \sigma\sqrt{t(1-t)}\epsilon$ with $t \sim \mathcal{U}[0, 1]$, $\epsilon \sim \mathcal{N}(0, I)$, and $\sigma = 0.02$; the label is the clean displacement $u = z_1^* - z_c$, and the perturbation, which vanishes at both endpoints, is absent at inference.

C. Band-Weighted Velocity Matching

The network $v_\theta(z_t, t, s, x_s)$ has a shared trunk of four width-64 residual blocks and separate 1×1 heads for LL, LH, and HL. With $(u_\ell, u_{h1}, u_{h2}, u_{h3}) = \mathcal{W}(u)$, it minimizes

$$\mathcal{L} = \mathbb{E}_t[\lambda_{LL}\|v_\ell - u_\ell\|^2 + \|v_{h1} - u_{h1}\|^2 + \|v_{h2} - u_{h2}\|^2], \quad (4)$$

with $\lambda_{LL} = 0.3$. The endpoint (2) changes the requested LL velocity, whereas λ_{LL} changes its optimization pressure. A style memory indexed by s gives the trunk a coarse prior. For image-specific texture, the reference LH and HL bands are pooled into coordinate-free, orientation-specific codes that modulate only the matching heads; feeding raw spatial reference bands instead leaks the reference layout (DINO-C falls to 0.40 in our probes; see supplement). HH has no learned head. The model has 1,037,087 trainable parameters.

D. Stepwise High-Frequency Alignment

Velocity matching organizes the motion but leaves a weak appearance signal. With $q = P_H z$, $q_s = P_H z_s$, and $\mathcal{A}(q; q_s) = \sigma(q_s)(q - \mu(q))/\sigma(q) + \mu(q_s)$ over channel-wise spatial statistics, each of the K inference steps is

$$\begin{aligned} \bar{z}^{k+1} &= z^k + \frac{1}{K} \mathcal{W}^{-1}(v_\ell^k, v_{h1}^k, v_{h2}^k, 0), \\ z^{k+1} &= P_L \bar{z}^{k+1} + (1 - \beta) \bar{q} + \beta \mathcal{A}(\bar{q}; (1 + \eta)q_s), \end{aligned} \quad (5)$$

where $\bar{q} = P_H \bar{z}^{k+1}$, $K = 8$, $\beta = 2$, and $\eta = 0.1$. With $\beta > 1$ the update over-relaxes toward the matched statistics; applied at every step, alignment becomes part of the trajectory rather than a post-process and supplies HH statistics without a learned head.

E. Metric-Free Stopping

Training needs no decoding, inversion, or discriminator and runs 52 steps per epoch at batch 96. After each epoch, a probe batch of four fixed latents at $t = 0.5$ gives ρ_e , the ratio of LL-loss to HF-loss gradient norms on the shared trunk. Training stops at the first epoch in which the gate of the texture route

TABLE I

MAIN COMPARISON (D-S: DINO-S, C-S: CLIP-S, LP: LPIPS, D-C: DINO-C). EXCLUDING IDT AND TGT, THE BEST ENTRY PER COLUMN IS **BOLD** AND THE SECOND UNDERLINED. †: STYLE NOT ABOVE IDT; ‡: CONTENT BEYOND TGT; VALID: BENCHMARKS WITHOUT EITHER MARK. PARAMS IN MILLIONS, FROZEN PARAMETERS AS SUPERScript; PARAMS AND TRAIN RANK TRAINED METHODS. WEAVE USES ITS SELECTED D5 CHECKPOINT ON D5 AND DATASET-SPECIFIC CHECKPOINTS ON P2A/R5; COST REFERS TO THE D5 MODEL. TIMES ON ONE RTX 3060; *: AUTHOR REPORT ON FOUR GPUS [16]; ~3 h: Z-STAR ESTIMATED FROM STEP-SCALED RUNS (FULL 512 PX RUNS EXCEED 12 GB).

Method	D5-512				P2A-256				R5-WikiArt				Params	Train	Infer	Valid
	D-S†	C-S†	LP↓	D-C†	D-S†	C-S†	LP↓	D-C†	D-S†	C-S†	LP↓	D-C†	(M)	(min)	(750)	boards
IDT (identity)	0.419	0.693	0.000	1.000	0.416	0.663	0.000	1.000	0.433	0.731	0.000	1.000	0	free	0 s	–
TGT (target style)	1.000	0.863	0.776	0.215	1.000	0.835	0.800	0.169	1.000	0.796	0.747	0.230	0	free	0 s	–
SD-Turbo	0.484	0.693 [†]	0.003	0.922	0.341 [†]	0.674	0.603	0.279	0.505	0.767	0.449	0.538	0 ⁺¹¹⁰⁰	free	5 m	1/3
StyleAligned	0.675	0.780	0.869 [‡]	0.239	0.612	0.768	0.786	0.310	0.649	0.824	0.829 [‡]	0.315	0 ⁺¹¹⁰⁰	free	77 m	1/3
Z-STAR	0.449	0.784	0.347	0.549	0.498	0.786	0.332	0.552	0.514	<u>0.822</u>	0.384	0.526	0 ⁺¹¹⁰⁰	free	~3 h	3/3
StyleShot	<u>0.563</u>	<u>0.787</u>	0.765	0.377	<u>0.552</u>	<u>0.774</u>	0.713	0.335	0.484	0.787	0.795 [‡]	0.380	0 ⁺²⁴⁰⁰	free	5.1 h	2/3
StyleID	0.548	0.822	0.552	0.416	0.256 [†]	0.648 [†]	0.691	0.147 [‡]	<u>0.551</u>	0.815	0.493	0.385	0 ⁺¹¹⁰⁰	free	63 m	2/3
CUT	0.471	0.714	0.374	0.795	0.539	0.754	0.494	0.702	0.441	0.710 [†]	0.620	<u>0.795</u>	<u>7.0</u>	322.6	5 m	2/3
SaMST	0.271 [†]	0.618 [†]	0.749	0.145 [‡]	0.501	0.709	0.398	0.838	0.461	0.667 [†]	0.612	0.615	8.3	<u>39.5</u>	10 m	1/3
SaMam	0.477	0.582 [†]	<u>0.243</u>	<u>0.812</u>	0.505	0.677	0.205	0.870	0.503	0.712 [†]	0.227	0.925	8.8	436.0	17.6 m	1/3
StyTr-2	0.501	0.709	<u>0.596</u>	<u>0.747</u>	0.515	0.722	0.495	0.555	0.533	0.759	0.553	0.737	35.4 ^{+12.9}	~1440*	38 m	3/3
AesPA-Net	0.520	0.724	0.548	0.601	0.498	0.744	0.571	0.522	0.545	0.782	0.409	0.728	11.3 ^{+20.0}	–	23 m	3/3
Seedream 4.5	0.486	0.720	0.477	0.739	0.517	0.752	<u>0.227</u>	0.785	0.504	0.742	0.486	0.725	–	API	–	3/3
Latent-WCT	0.362 [†]	0.673 [†]	0.441	0.559	0.338 [†]	0.738	0.585	0.386	0.414 [†]	0.710 [†]	0.444	0.553	0	free	18 s	0/3
WEAVE (ours)	0.492	0.713	0.260	0.810	0.480	0.668	0.312	<u>0.861</u>	0.523	0.775	<u>0.289</u>	0.772	1.04 ⁺⁸⁴	1.4	<u>126 s</u>	3/3

grows and $\rho_e/\rho_{e-1} \leq 0.65$, i.e., when high-frequency learning takes over the trunk. On D5 the rule fires at epoch 4 (ρ drops from 1.09 to 0.16) after 82.8 s, without any image metric.

V. EXPERIMENTS

A. Setup

Benchmarks. *D5-512*: five WikiArt styles (Early Renaissance, Impressionism, Minimalism, Rococo, Ukiyo-e) at 512 px with 3,600 training images each; 30 held-out sources per style are sent to all five styles (750 requests, same-style requests kept). *P2A-256*: the same layout over photo, Monet, Van Gogh, Cézanne, and Hayao at 256 px. *R5-WikiArt*: five randomly drawn WikiArt families (Cubism, Expressionism, Pop Art, Romanticism, Symbolism) at 512 px.

Metrics. DINO-S (primary) is the maximum cosine similarity between the DINOv2-small CLS feature [44] of the output and those of up to 30 held-out references of the target style; it rewards resembling some member of the family rather than its average. CLIP-S is the CLIP ViT-B/32 [45] similarity to the target-style prototype, DINO-C the DINOv2 similarity to the source, and LPIPS uses AlexNet.

Baselines and implementation. We compare training-free diffusion editors (SD-Turbo [32], StyleAligned, Z-STAR, StyleShot, StyleID), learned models (CUT, SaMST, SaMam, StyTr², AesPA-Net), the Seedream 4.5 API [46], and an analytic Latent-WCT, each at its released or recommended operating point. WEAVE uses AdamW (learning rate 2×10^{-4} , cosine decay), bf16, and one RTX 3060 (12 GB). Training time excludes one-off latent caching; inference time includes eight ODE steps and VAE decoding.

B. Main Results

The audit. Eight of twelve baselines fail the sandwich on at least one benchmark (Table I, Fig. 1b). SD-Turbo’s best-in-class D5 LPIPS (0.003) is a near-perfect copy: its CLIP-S equals that of IDT. SaMam, the strongest compact baseline on content, falls below IDT in CLIP-S on two benchmarks, so its low LPIPS comes at the price of style. StyleAligned tops DINO-S everywhere but drifts farther from the source than the reference image itself on D5 and R5. Latent-WCT never clears IDT: wavelet statistics alone do not transfer style; the learned transport does.

Valid methods. Among the five methods valid everywhere (Z-STAR, StyTr², AesPA-Net, Seedream 4.5, and WEAVE), WEAVE has the highest DINO-C on all three benchmarks (0.810/0.861/0.772) and the lowest LPIPS on D5 and R5. It surpasses the commercial Seedream 4.5 in DINO-S on D5 (0.492 vs. 0.486) at about half its LPIPS (0.260 vs. 0.477). At 512 px, no method, valid or not, beats WEAVE on a style and a content metric at once: WEAVE is Pareto-optimal against all twelve baselines in every style–content pair on both D5 and R5.

Cost. WEAVE trains $28\times$ faster than the fastest learned baseline and over $300\times$ faster than SaMam (1.4 vs. 39.5 and 436 min), runs at 168 ms per 512 px image including VAE decoding, and is $11\times$ smaller and at least $11\times$ faster at inference than every other valid learned model.

Qualitative comparison. Fig. 4 shows both failure modes on an extreme request, and the close-ups show what the metrics measure: WEAVE keeps the geometry of the source and rewrites its texture and palette.

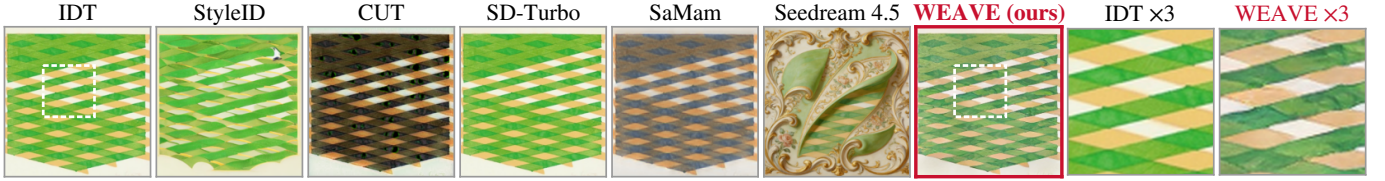


Fig. 4. Extreme request (Minimalism $\rightarrow$ Rococo). SD-Turbo returns the input (identity shortcut); Seedream redraws the scene with new ornaments. WEAVE keeps the lattice and, in the close-ups (dashed boxes, 3 $\times$), turns the flat Minimalist fill into varied, painterly strokes in a warmer palette.

TABLE II
COMPONENT ABLATION ON D5-512 (750 REQUESTS).

Variant	DINO-S $\uparrow$	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-C $\uparrow$
WEAVE (epoch 4)	0.4918	0.7128	0.2595	0.8102
w/o stepwise AdaIN ($\beta=0$)	0.4821	0.7005	0.2258	0.8455
w/o texture conditioning	0.4825	0.7157	0.2837	0.8125
direct endpoint ($z_1^* = z_s$)	0.4894	0.7172	0.3138	0.7966
$\lambda_{LL}=1.0$	0.4910	0.7180	0.2626	0.7874
learned HH head	0.4930	0.7164	0.2670	0.8061

TABLE III
BEYOND THE TRAINING SETTING. TOP: ZERO-SHOT ON FIVE UNSEEN WIKIART FAMILIES (750 PAIRS, D5 CHECKPOINTS). BOTTOM: HAAR ALIGNMENT IN A FROZEN SD1.5 EDITOR (600 PAIRS).

Method	DINO-S $\uparrow$	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-C $\uparrow$
SaMST	0.3913	0.7485	0.6213	0.2680
SaMam	0.5466	0.7577	0.3990	0.7369
WEAVE	0.5313	0.7742	0.2802	0.7587
SD1.5 img2img	0.3974	0.7221	0.3591	0.6589
+ Haar alignment	0.4080	0.7366	0.3630	0.6531

C. Ablations

Table II removes one component at a time on the same 750 requests; inference-time controls reuse the selected checkpoint, and retrained variants (seed 42) are reported at their best-DINO-S epoch, which favors them. Every component we keep pays for itself (seed s.d. of DINO-S: 0.003). Without stepwise AdaIN the output slides back toward the source (DINO-S -0.010 , CLIP-S -0.012). Without texture conditioning DINO-S drops by 0.009 and LPIPS rises by 0.024. Replacing the anchored endpoint by the direct target costs the most content (LPIPS $+0.054$, DINO-C -0.014), exactly as Proposition 1 predicts, and delays the style peak by nine epochs: the anchor is what makes 82-second training possible. Raising λ_{LL} to 1.0 leaves DINO-S unchanged but costs 0.023 DINO-C, and a learned HH head adds content cost (LPIPS $+0.008$) for a marginal style gain, so WEAVE omits it.

D. Robustness and Transfer

Seeds and stopping. Without any image metric in the loop, the stopping rule picks epochs 4, 4, and 3 for seeds 42, 7, and 123 (DINO-S 0.490 ± 0.003 , LPIPS 0.261 ± 0.006) and hits the DINO-S peak exactly on both seeds with full 15-epoch curves.

Reference pools. Redrawing the DINO-S reference pool 1,000 times with 8 of the 30 references per style keeps the margin over IDT positive in every draw (0.030, 95% interval $[0.020, 0.039]$), essentially equal to the full-pool margin (0.033): the gain does not hinge on any particular reference, including the one WEAVE conditions on.

Unseen families. The D5 checkpoint transfers zero-shot to five held-out WikiArt families with the best CLIP-S, LPIPS, and DINO-C (Table III, top); SaMam’s 0.015 DINO-S edge costs it 0.12 LPIPS.

Plug-in for a frozen editor. The same Haar-domain alignment (LH/HL/HH at scale 0.8, LL untouched) upgrades a frozen

SD1.5 image-to-image pipeline (empty prompt, strength 0.5, 20 DDIM steps) without retraining: on 600 cross-style D5 pairs, DINO-S and CLIP-S rise significantly (paired t -test, $p < 10^{-15}$) for $+0.004$ LPIPS and 4.9 ms per image (Table III, bottom).

VI. CONCLUSION

Standard metrics reward copying, and two zero-cost references are enough to catch it: under them, eight of twelve baselines fail, and every compact one falls below the identity in style on some benchmark. The cause is spectral, and WEAVE removes it at the source: it anchors structure, learns texture transport in explicit subbands, and injects high-frequency statistics along the trajectory. The result is the only compact model that is valid everywhere and Pareto-optimal at 512 px, trained in 1.4 minutes with 1.04M parameters. Multi-scale decompositions and other autoencoders are natural next steps. Code, checkpoints, and the 750-request protocol will be released.

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